ASSIGNMENT

1. Find the singular value decomposition of $A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \\ 1 & 1 \end{bmatrix}$.
2. Find the basis and dimension of $W = \left\{ \begin{bmatrix} s+t \\ 2s \\ 3s-t \\ -t \end{bmatrix}; s,t \in \mathcal{R} \right\}$
3. Find the homogeneous system whose solution space is spanned by the set $S = \{(1,1,2,1,1), (1,2,1,4,3), (3,5,4,9,7)\}$
4. Find the basis and dimension of the null space of given homogeneous system of equations

$$x + 2y - 2z + 2s - t = 0$$

$$x + 2y - z + 3s - 2t = 0$$

$$2x + 4y - 7z + s + t = 0$$